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Project Management 4.2

Quality in Project Management

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Learning Outcomes

- By the end of this week, you will be able to:
  - State why quality management is important
  - Discuss various methods to manage quality
  - Identify project risks to quality
  - Give relevant quality processes to ensure quality is the highest possible.

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Planning Continued

- What is Quality?
- Quality is essential to plan in advance
- Quality does not happen automatically

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Quality in a Project Environment

- Quality is often used to describe a resource (along with time and cost) we use to deploy our project
- But it is also used to describe the project environment in which we work;
  - It is not enough to deploy the project, we need to follow any prescribed processes
  - It is not enough to be successful, we need to know why so that we can repeat that success.

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Quality Management

- Quality Management is a process
- Quality Management Plan
- There are a number of elements to quality management

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Quality Management (Continued)

- Quality Assurance
- Quality Planning
- Quality Control

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Quality in a Project Environment

- Quality is a phenomenon
  - It can be hard to describe in terms everyone will agree upon, e.g.
  - Be very specific

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Quality in a Project Environment

- Purpose
  - Meet Business Expectations
  - Enable the Desired Benefits to be Achieved
- Product/Service Focussed
- Understanding what will be Produced.

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The Need for Quality – IT Example

- I.T. Companies or I.T. Departments must produce quality Software / Systems
- Quality means as few faults in Software / Systems as possible
- Increased effort and cost is incurred in finding faults later in the development cycle
- Quality procedures and culture can find faults earlier
- Quality must be designed into the project, and its plans.

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Quality

- However you use quality
  - In describing resources available to the project
  - In describing how the project will be deployed
  - In defining what the project will deliver
- Be sure that stakeholders share a common understanding of what quality means in order to avoid disputes, rework, change and dissatisfaction that could affect the perceived or actual success of your project.

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Organisation and Culture

An Organisation will not successfully incorporate Quality into its products without first providing a Quality Culture.

- People are as important as tools or procedures
- Staff acceptance is vital
- QMS seen as new working practices, benefits staff
- Staff motivation needs Top Management commitment
- A Quality Culture needs time to develop
- Quality must be planned into an Organisation.

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Managing Quality Assurance

A Project Manager should address the following issues if they are to successfully manage the quality of a project:

- Planning for Quality, Quality Plan
- Team Organisation, Auditors, Coordinators
- Quality Procedures and their enforcements
- Quality Training – Project Team Meetings
- Methodologies
- Documented development standards
- Standards.

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Managing Quality Assurance

- Measurement and record keeping, audits, test results
- Quality Control, Signoffs, Walk through
- Development Documentation, Forms, Telephone Logs, Test Plans,
- Software testing
- Technical Reviews
- Managing User Involvement, Change Control.
- Full system documentation.

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Planning Quality

- Quality must be planned
- Project Quality Plan
- The Stage Plan
- Generic plan
- Defines how quality is to be monitored
- Responsibilities for stages
- It may include specifics, response times etc.
- Often an independent quality assurance function to monitor and comment on Project Quality Plans.

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Quality Co-ordination

One way to co-ordinate quality on a project is to appoint a Quality Coordinator

- A System Analyst often performs this role
- Responsible to I.T. Manager or Project Manager
- Attends management meetings across projects
- Controls and keeps up to date the Company QMS
- Point of contact for staff to suggest changes
- Attends Walkthroughs, all stages of development
- Ensures new staff are trained, aware of procedures
- Provides information, news bulletins, staff informed
- Encourages the growth of a Quality Culture.

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Quality Circles

- Gillies, A.C. (1992) is a useful source of information regarding Quality Circles
- Quality Circles can be used to get staff involved
- Quality Circle, 3-15 people, Meets every 1-3 weeks
- Issues of Quality and Improvements discussed
- Leaders trained, all staff must understand why quality is important.

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Document Review Planning

- When to Walkthrough:
  - The author must believe the document is complete, there can be more than one W/T. e.g. rework, large documents, W/T in parts; e.g. contents page
- Planning:
  - Issues to consider, risk assessment, what must/should be W/T, associated documents, who should attend, are signoffs required, has time been budgeted in the plan
- Document Issue and Preparation:
  - Allow staff time to study the document before W/T, individuals should focus on different areas.

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Document Review Planning

- Who is involved?
  - Between 3-6 people depending on size/complexity
  - Invite correct people, e.g. Design Document
- The Roles:
  - Author, Moderator, Scribe, Quality Co-ordinator, User,
  - Business/Technical Experts, Naïve User, Next Author.
- Documentation used
  - Form for every W/T held, records points raised,
  - Acts as a Quality Control Record,
  - Proof of QA checks, aids Rework.

Review Documentation

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Project: _____ **Attendees** **Role**

Document: _____ _____ _____

Author: _____ _____ _____

Date of Review: _____ _____ _____

Ref	Description	Code	Action	Complete

Decision

Accept/Rework/Rewrite

Comments:

Moderator's Signature:

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Document Review Guidelines

- Do Find as many faults /improvements as possible
- Do Share equally in the inspection
- Do Question areas that are not fully understood
- Don’t Spend more than 2 hours in one walkthrough
- Don’t Apportion blame or guilt
- Don’t Criticise or lecture the author.

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Document Review Guidelines (Continued)

- Don’t Look for solutions. This is done afterwards
- Don’t Discuss points. Wastes fault finding time
- Don’t Inhibit by discussing whether point is valid
- Don’t miss suggestions triggering new ideas
- Ending the Walkthrough
  - Accept, Rework, New Walkthrough, Signoff.

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Roles and Responsibilities

- Project teams vary
- Responsibilities
  - As the project team is a temporary structure individuals may be unfamiliar with their role at the beginning and
  - Those outside of the project need to know who does what.

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Quality Register (PRINCE2)

- The Quality Register is effectively a diary of the quality events planned and undertaken (for example, workshops, reviews, inspections, testing, pilots, acceptance and audits).
- It is created during the Initiating a Project process as the products and quality control measures are being defined.
- It is then maintained throughout the project.

Quality Register

- Updated activities.
- Key audit and assurance
- Information

Quality Register Example

Products:	

Reference Number	Product	Method of Quality Checking	Staff Responsible /Role	Planned Date	Actual Date	Result	Number of Action Items	Target Sign-off Date	Actual Sign-off Date

http://p2weasel.xtreemhost.com/html/appendices/product_description.html#q

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Quality Regardless

- Every environment will approach quality differently
- There is a difference between the quality of the project management, and the quality of project outputs
- They should BOTH be of the highest quality.

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Look Forward

- Project learning.